

Lecture 3

1/26/26
JRA

Lecture overview $\left\{ \begin{array}{l} \text{Ray tracing} \\ \text{Lens equations} \end{array} \right. \left| \begin{array}{l} \text{Formalisms for} \\ \text{finding image} \\ \text{positions, magnification,} \\ \text{etc.} \end{array} \right.$

Ray Tracing

All rays from a given point in object come to focus at same point in image (ignoring aberrations)

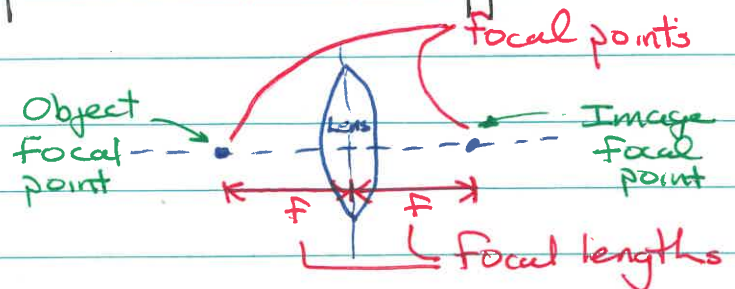
Ray tracing uses three particular rays that are easy to deduce & draw to find image location, image size, & magnification. Any two of these rays are sufficient but customary to draw all three as a double check. See Tutorial in textbook.

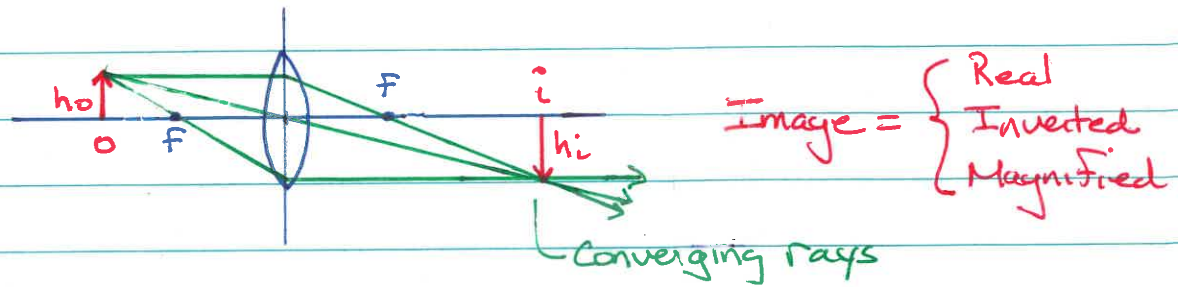
Look at converging (bulging) lenses.

Small bulge $\Rightarrow$ less lens power

Large bulge $\Rightarrow$ more lens power

Lenses have two "focal points" symmetrically positioned at a distance " f ," called the "focal length," along the optical axis on opposite sides of the lens





Reviewed ray tracings on PP slide ("Attributes of a Converging Lens") w/ three different object positions.

Decreasing $o \Rightarrow \begin{cases} \text{Increasing } i & (\text{image distance}) \\ \text{Increasing } h_i & (\text{image size}) \end{cases}$

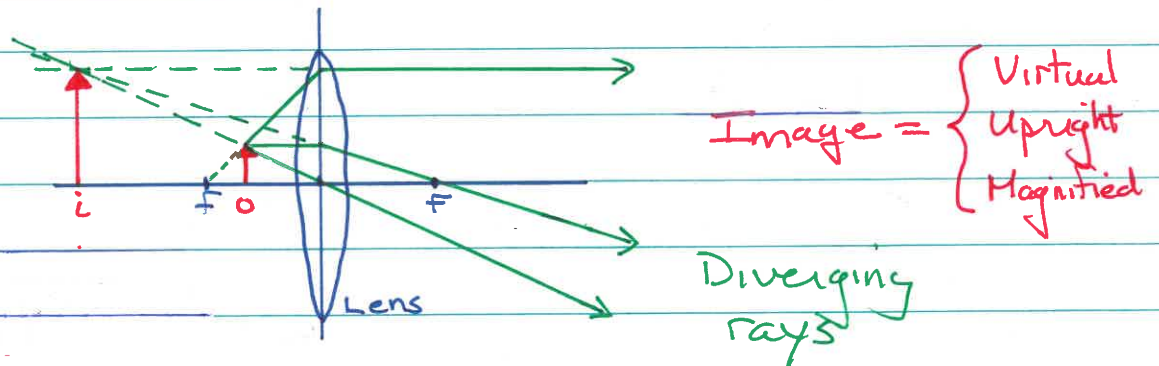
Discussed "Real images" & "Virtual images"

Real Image: • light is actually brought to focus
• image has energy to excite camera pixels & burn a piece of paper

Real images are only formed if the object distance, o , is greater than the focal length, f .

Virtual Image: • light is NOT brought to focus
• no energy at position of image

Virtual images are formed if the object distance is less than the focal length. In this case, the lens is NOT powerful enough to bring the rays to focus. Upshot: no real image.



Can use another converging lens, like an eye, to bring the diverging rays to focus to form a real image, like on a retina.

The eye interprets the diverging rays as coming from a "virtual image" positioned at a location upstream from the object where there is no actual light energy.

Always get an upright virtual magnified image when object positioned within focal length of lens.

This class will NOT look at diverging lenses (except in a HW problem).

Lens Equations

Lens equations are a mathematical complement to ray tracing. They give the same answers.

Thin Lens Eq

$$\frac{1}{i} + \frac{1}{o} = \frac{1}{f}$$

HW - prove thin-lens eqn using a similar triangles argument

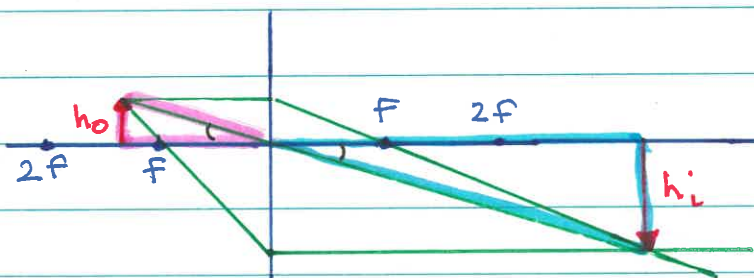
The thin-lens eqn can be used to solve for image position given object position & focal length (among other possibilities, e.g., calculate F given i & o).

Magnification can be defined as a ratio of image height to object height & image position to object position.

$$\text{Magnification} = m = \frac{\text{image height}}{\text{object height}} = \frac{h_i}{h_o} = -\frac{i}{o}$$

Minus sign for inverted image

Thus, a knowledge of i & o from the thin-lens eqn allows calculation of magnification



△ and ▽ in the drawing above are similar triangles, so angles are the same, & ratios of similar side lengths in the two triangles are the same

$$\frac{h_i}{h_o} = -\frac{i}{o}$$

The same result can be obtained by noting that the angles opposite the two heights in the two triangles are the same.

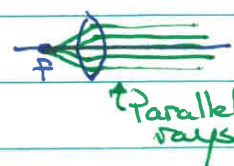
$$\tan \theta = \frac{h_o}{o} = \frac{h_i}{i} \Rightarrow \frac{h_i}{h_o} = \frac{i}{o}$$

Fill out chart showing characteristics of image, exemplary applications, etc. based on object position relative to focal length

Object Distance	Image Type	Image Distance	Image Orientation	Image Size	Application(s)
$\infty > o > 2f$	R	$f < i < 2f$	Inverted	Minified	eye, camera
$o = 2f$	R	$i = 2f$	Inverted	Mag = 1	photocopier
$2f > o > f$	R	$2f < i < \infty$	Inverted	Magnified	microscope objective
$o = f$	$\emptyset$	No Focus Light parallel	No IMAGE	$\emptyset$	collimator
$o < f$	V	$ i > o$	Upright	Magnified	microscope eyepiece

↑ image to left of object

Collimators use $o = f$ to collimate light



ACTIVITY 1

Virtual Image Formation

See separate Activity page for solution

Name Key**Physics 390 In-Class Activity 1*****Virtual Image Formation***

A light bulb is 0.75 m from a converging lens with $f = 1.00$ m.

(a) Compute the image distance and magnification using appropriate equations.

$$1/u + 1/o = 1/f$$

$$1/u = 1/f - 1/o = \frac{o-f}{of}$$

$$M = -u/o = 3/0.75 = 4$$

$$u = \frac{of}{o-f} = \frac{0.75}{-0.25} = -3\text{m}$$

(b) Repeat your analysis by drawing a ray diagram on the graph paper below. Make your object fairly small because part (a) predicts significant magnification.

